Fine-tuning and guardrails

AI agents · University · Lesson 4



In this lesson you will

- Measure what fine-tuning teaches and what it makes a model forget.
- Explain why learned safety is not a guarantee.
- Design layered guardrails in code: tools, approval, checks, logs, tests.

Key ideas

- Fine-tuning teaches and forgets: with no replay, 7/8 new but 1/8 old.
- Learned refusals and asking first can fail: 28% and 12% of reworded refusals got a call.
- Guard actions in code: fewest tools, exact approval, checks, logs, re-tests.

Activity: Layers of defence

1. Pick (3 min): An agent and one risky action.
2. Layers (7 min): Write five guards, one per card.
3. Attack (7 min): A partner removes cards one at a time.
4. Debrief (3 min): Which guard mattered most?

Use made-up agents, people and messages only. Five index cards or torn paper squares are enough.

Let’s discuss

1. When is fine-tuning worth it, and when is a new tool or a clearer instruction enough?
2. Why is a model’s promise to ask first not enough?
3. Which guardrail would you never remove, and why?

Exit ticket

- One thing fine-tuning can make a model forget, and the guard in code that still protects you.